THE ANATOMY OF A VIRUS

A biological virus is a very small, simple organism that infects living cells, known as the host, by attaching itself to them and using them to reproduce itself. This often causes harms to the host cells.

Similarly, a computer virus is a very small program, routine that infects a computer system and uses its resources to reproduce itself. It often does this by patching the operating system to enable it to detect program files, **such as** COM or EXE files. It then copies itself into those files. This sometimes causes harm to the host computer system.

When the user runs an infected program, it is loaded into memory carrying the virus. The virus uses a common programming technique to stay resident in memory. It can then use a reproduction routine to infect other programs. This process continues until the computer is switched off.

This virus may **also** contain a payload that remains dormant until a trigger event activates it, such as the user pressing a particular key. The payload can have a variety of forms. It might do something relatively harmless such as displaying a message on the monitor screen, or it might do something more destructive such as deleting files on the hard disk.

When it infects a file, the virus replaces the first instruction in the host program with a command that changes the normal execution sequence. This type of command is known as a JUMP command and causes the virus instructions to be executed before the host program. The virus then returns control to the host program which then continues with its normal sequence of instructions and is executed in the normal way.

To be a virus, a program only needs to have a reproduction routine that enables it to infect other programs. Viruses can, however, have four main parts. A misdirection routine that enables it to hide itself; a reproduction routine that allows it to copy itself to other programs; a trigger that causes the payload to be activated at a particular time or when a particular event takes place; and a payload that may be a fairly harmless joke or may be very destructive. A program that has a payload **but** does not have a reproduction routine is known as a Trojan.

Ejercicios

1. Leer el texto y encontrar las respuestas a las siguientes preguntas. Las respuestas deben ser en español, no en inglés

1) How are computer viruses like biological viruses?
2) What is the effect of a virus patching the operating system?
3) Why are some viruses designed to be loaded into memory?

4) What examples of payload does the writer of the text provide?
5) How does a Trojan differ from a virus?

2. Unir cada rutina de virus con su función:

Routine:

I. *Misdirection*
II. *Reproduction*
III. *Trigger*
IV. *Payload*

Function:

a) *Does the damage*
b) *Attaches a copy of itself to another program*
c) *Hides the presence of the code*
d) *Decides when and how to activate the payload*

3. Conectores. Completar el siguiente cuadro en español.

Conector	**Traducción**	**Tipo de relación que establece**	**Idea 1**	**Idea 2**
Such as				
Also				
To				

But				

4. Referencia contextual. Encontrar los referentes a las siguientes palabras.

- Itself……
- Them……
- Its ……
- It……
- Which ……

5. Traducir las siguientes frases:

-host computer system

-common programming techniques

-normal sequence of instructions

-harmless joke

-deleting files on the hard disk

RESUELTO

Práctica de clase · The anatomy of a virus

Las respuestas van en azul. Las consignas son las mismas de las hojas anteriores.

1. Leer el texto y encontrar las respuestas a las siguientes preguntas (en español).

• How are computer viruses like biological viruses?
→ Los dos son muy pequeños, infectan a un huésped (una célula o un sistema informático), usan sus recursos para reproducirse y muchas veces le causan daño.

• What is the effect of a virus patching the operating system?
→ Le permite detectar los archivos de programa (como los COM o EXE) para copiarse dentro de ellos.

• Why are some viruses designed to be loaded into memory?
→ Para quedar residentes en memoria y, desde ahí, infectar otros programas hasta que se apague la computadora.

• What examples of payload does the writer of the text provide?
→ Uno inofensivo: mostrar un mensaje en el monitor. Uno destructivo: borrar archivos del disco rígido.

• How does a Trojan differ from a virus?
→ El troyano tiene payload, pero no tiene rutina de reproducción: no se copia a otros programas.

2. Unir cada rutina de virus con su función.

• I. Misdirection
→ c) Hides the presence of the code

• II. Reproduction
→ b) Attaches a copy of itself to another program

• III. Trigger
→ d) Decides when and how to activate the payload

• IV. Payload
→ a) Does the damage

3. Conectores. Completar el siguiente cuadro en español.

Conector	Traducción	Tipo de relación	Idea uno	Idea dos
Such as	como / tales como	ejemplificación	Archivos de programa.	Archivos COM o EXE.
Also	también	adición	El virus se queda en memoria e infecta otros programas.	Puede contener un payload.
To	para	finalidad	Ser un virus.	Un programa solo necesita una rutina de reproducción.
But	pero	contraste	El programa tiene un payload.	No tiene rutina de reproducción.

4. Referencia contextual. Encontrar los referentes a las siguientes palabras.

• Itself
→ a biological virus (el virus biológico)

• Them
→ living cells (las células vivas, el huésped)

• Its
→ a computer system (el sistema informático)

• It
→ an infected program (el programa infectado)

• Which
→ the host program (el programa huésped)

5. Traducir las siguientes frases.

• host computer system
→ sistema informático huésped (anfitrión)

• common programming techniques
→ técnicas de programación comunes

• normal sequence of instructions
→ secuencia normal de instrucciones

• harmless joke
→ broma inofensiva

• deleting files on the hard disk
→ borrar archivos del disco rígido